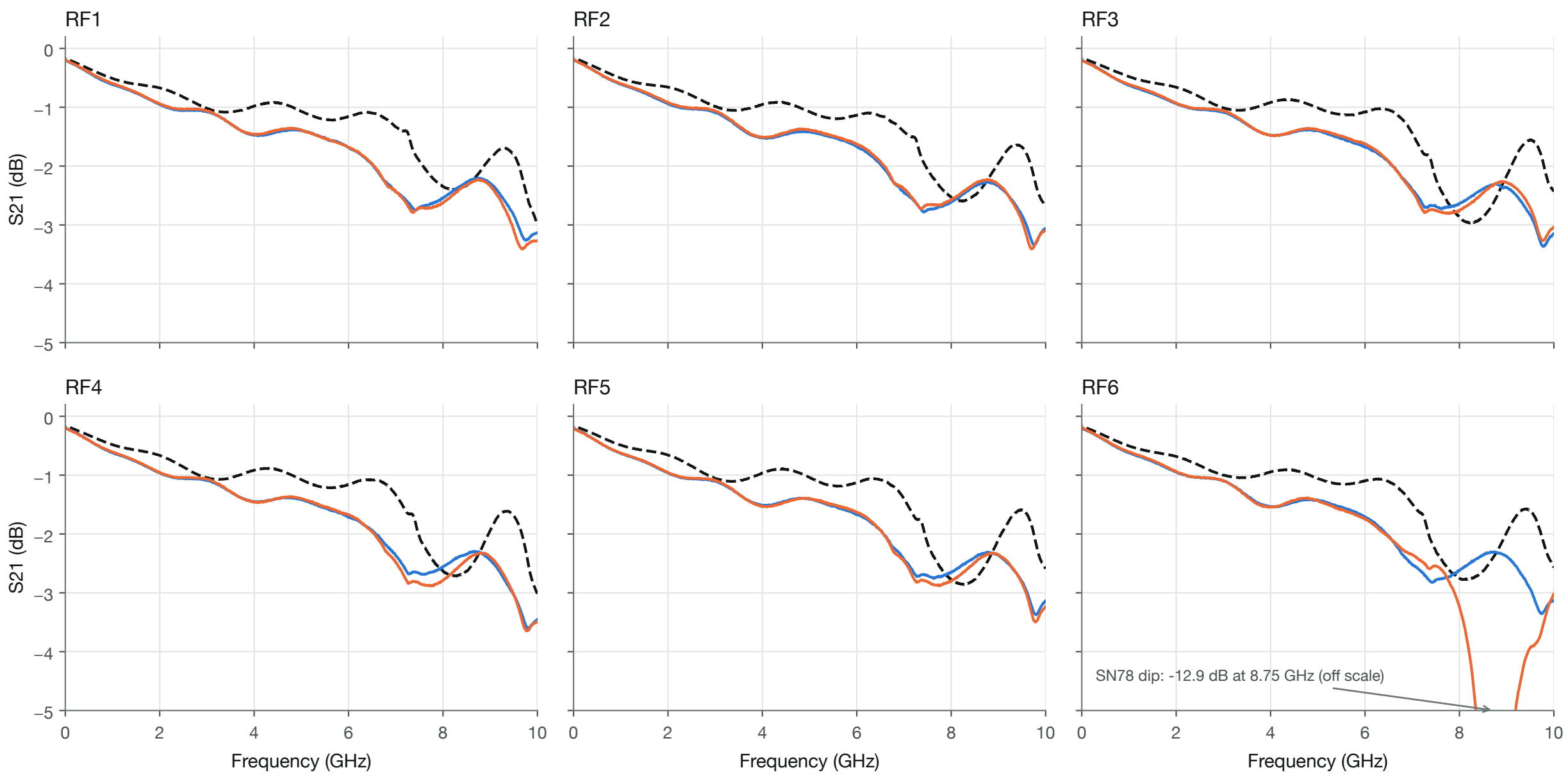


S21: measured switches vs calibrated NIST reference (295 K) — SN77 (measured) — SN78 (measured) - - - NIST SN0031 (calibrated)

NIST curve is SN0031's throw path only; measured curves also include the RFC connector section

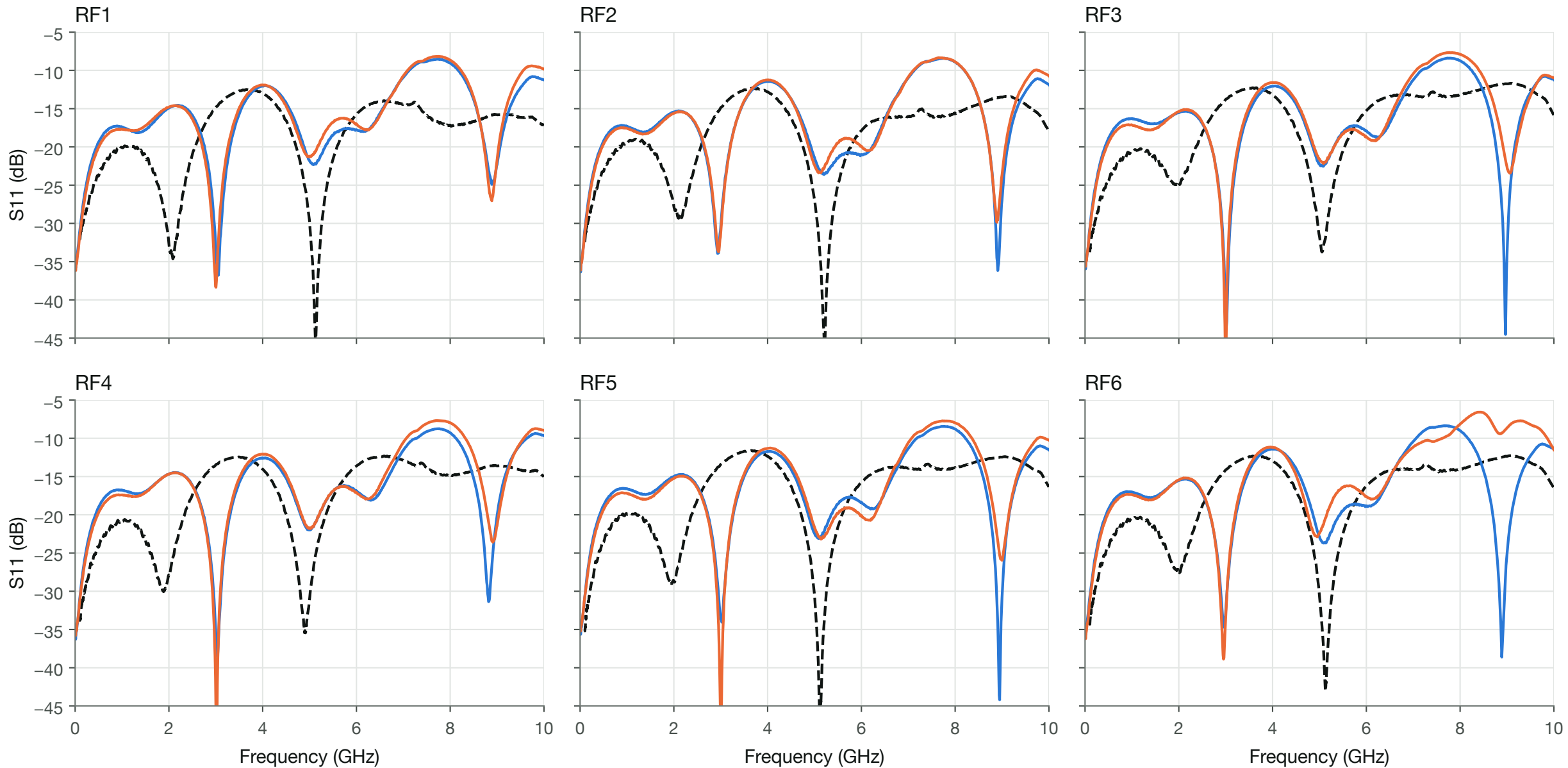


Measured: MM4250 SN77/SN78, 295 K, 11 Sep 2026, VNA-calibrated at switch connectors (RFC to RF port). NIST: SN0031 throw path from tier-2 cal, 295 K (internal junction to RF port; excludes RFC section).

S11: measured switches vs calibrated NIST reference (295 K)

NIST curve is SN0031's throw path only; measured curves also include the RFC connector section

— SN77 (measured) — SN78 (measured) - - - NIST SN0031 (calibrated)



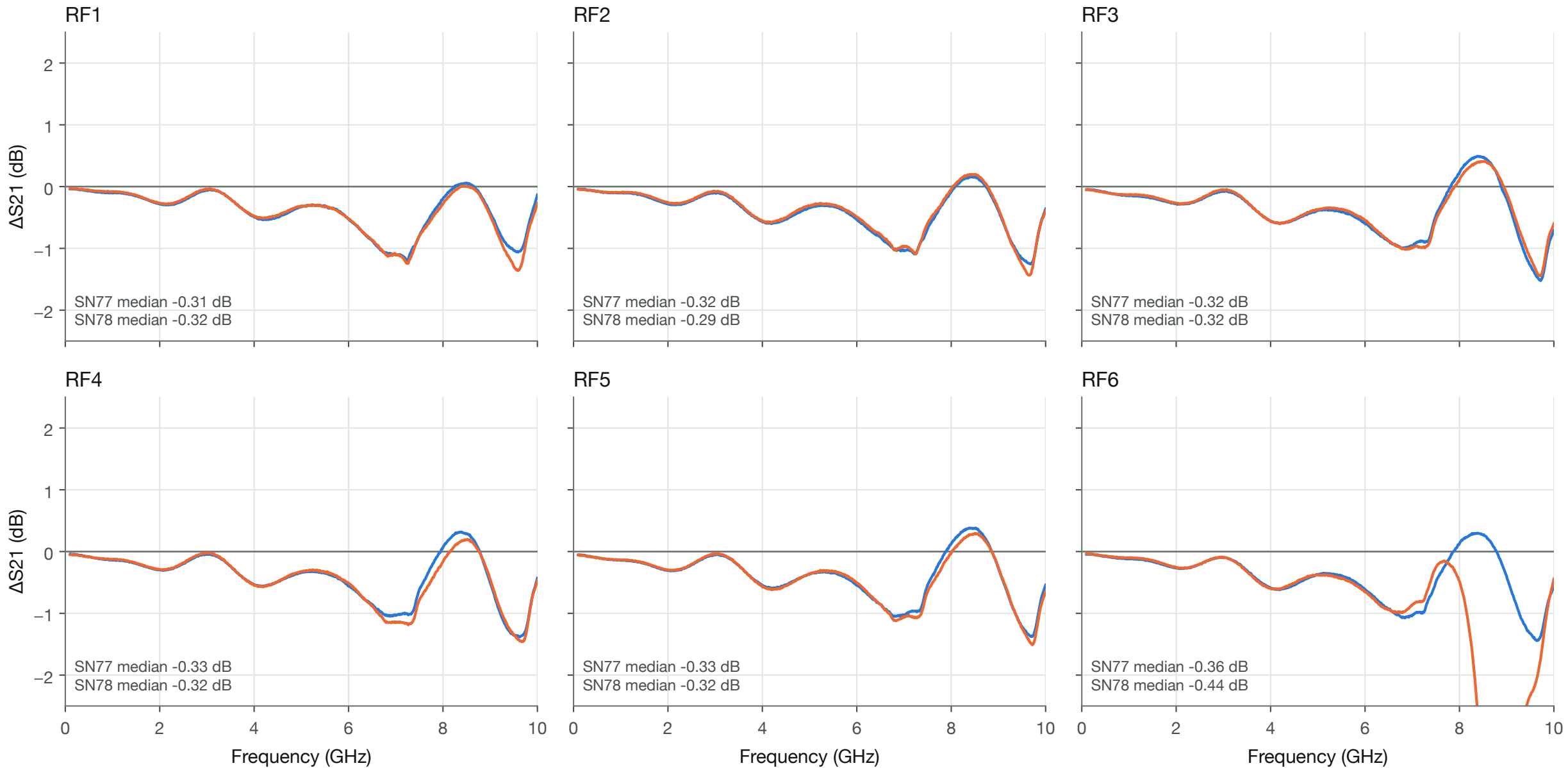
Measured: MM4250 SN77/SN78, 295 K, 11 Sep 2026, VNA-calibrated at switch connectors (RFC to RF port). NIST: SN0031 throw path from tier-2 cal, 295 K (internal junction to RF port; excludes RFC section).

S21 difference: measured minus calibrated NIST (295 K)

Below 0 = more loss than NIST. Part of the offset is the RFC connector section NIST's curve excludes.

— SN77 minus NIST

— SN78 minus NIST



Measured: MM4250 SN77/SN78, 295 K, 11 Sep 2026, VNA-calibrated at switch connectors (RFC to RF port). NIST: SN0031 throw path from tier-2 cal, 295 K (internal junction to RF port; excludes RFC section).

S11 difference: measured minus calibrated NIST (295 K)

Above 0 = closer to 0 dB (more reflection) than NIST.
Sharp spikes are where nulls fall at different frequencies (different reference planes); read the overall level, not individual spikes.

SN77 minus NIST SN78 minus NIST

